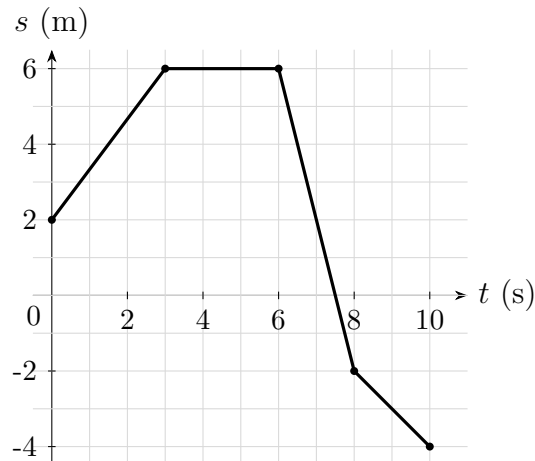


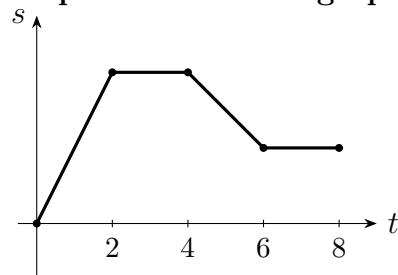
5.3 Classwork: Interpreting Motion Graphs

1. The displacement s (metres) of a particle from the origin is shown for $0 \leq t \leq 10$.

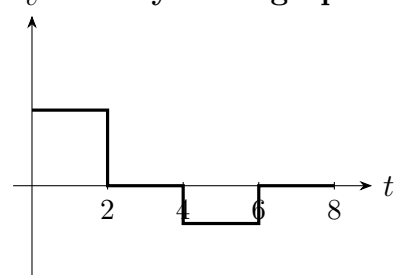


- Write down the initial displacement of the particle. [1]
 - Find the velocity during $0 \leq t \leq 3$. [2]
 - Describe the motion of the particle during $3 \leq t \leq 6$. [1]
 - Find the velocity during $6 \leq t \leq 8$. [2]
 - At what time does the particle pass through the origin? [2]
 - Find the total **distance** travelled from $t = 0$ to $t = 10$. [3]
 - Find the **displacement** from $t = 0$ to $t = 10$. [1]
2. The graphs below show the displacement and velocity of a particle. Match each description (i)–(iv) to the correct time interval. [4]

Displacement–time graph

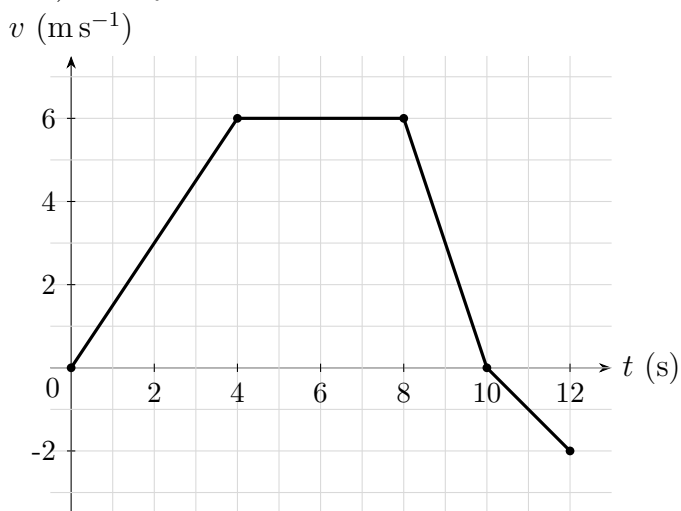


Velocity–time graph



- The particle is moving forward at constant speed.
- The particle is stationary.
- The particle is moving backward at constant speed.
- The particle is stationary (for the second time).

3. The velocity v (m s^{-1}) of a cyclist is shown for $0 \leq t \leq 12$ seconds.



- (a) Find the acceleration during $0 \leq t \leq 4$. [2]

- (b) Describe the motion during $4 \leq t \leq 8$. [1]

- (c) Find the acceleration during $8 \leq t \leq 10$. [2]

- (d) Find the total distance travelled from $t = 0$ to $t = 10$.

Hint: the distance is the area under the velocity curve. [3]

- (e) The cyclist continues with constant acceleration after $t = 10$. What does the negative velocity during $10 < t \leq 12$ tell you about the cyclist's motion? [1]

- (f) Find the total distance travelled from $t = 0$ to $t = 12$. [2]

- (g) Find the average speed of the cyclist over the 12-second interval. [2]